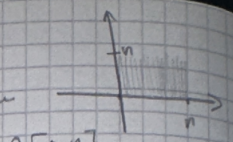
HW: Labs 3, 31.10.26

1 c.2) $(\mathbb{R}, \mathcal{B}_{\mathbb{R}}, \lambda)$

c) $f(\omega) := \begin{cases} n; & \omega \in \mathbb{Q}^c \cap [0,n] \\ 0; & \text{elsewhere} \end{cases}$ simple function



$$\int f \, d\lambda = n \, \lambda(\mathbb{Q}^c \cap [0,n]) + 0 \cdot \underbrace{\lambda(\mathbb{R} \setminus (\mathbb{Q}^c \cap [0,n]))}_{0}$$

$$= n \, \lambda(\underbrace{\mathbb{Q}^c \cap [0,n]}_{\text{irrationals on } [0,n]}) = n \cdot n = \underline{\underline{n^2}}$$

$$n - \lambda([0,n]) = \lambda\big((\mathbb{Q} \cup \mathbb{Q}^c) \cap [0,n]\big) =$$

$$= \lambda((\mathbb{Q} \cap [0,n]) \cup (\mathbb{Q}^c \cap [0,n])) =$$

$$= \lambda(\underbrace{\mathbb{Q} \cap [0,n]}_{\text{countable}}) + \lambda(\mathbb{Q}^c \cap [0,n])$$

Hence: $\lambda(\mathbb{Q}^c \cap [0,n]) = n$

Theory: Proposition 2.4.1

Let $\mathcal{C}$ be the set of all intervals of the form $(-\infty, a]$ in $\mathbb{R}$. Then $\sigma(\mathcal{C}) = \mathcal{B}_{\mathbb{R}}$.

Proof:

1) $\sigma(\mathcal{C}) \overset{?}{\subseteq} \mathcal{B}_{\mathbb{R}}$

$\mathcal{C}$) $=$

By definition $\mathcal{B}_{\mathbb{R}} = \sigma(\{(a,b) \mid a,b \in \mathbb{R}\})$ We also know that $\forall x \in \mathbb{R}: \{x\} \in \mathcal{B}_{\mathbb{R}}$ (lectures)

Then $\forall a \in \mathbb{R}: (-\infty, a] = (-\infty, a) \cup \{a\} \in \mathcal{B}_{\mathbb{R}} \Rightarrow \mathcal{C} \subseteq \mathcal{B}_{\mathbb{R}}$ (lectures)

By definition of the generated σ-algebra:

$\sigma(\mathcal{C}) := \bigcap_{i \in I} \mathcal{F}_i$, $\mathcal{C} \subseteq \mathcal{F}_i$. Then $\exists j \in I: \mathcal{F}_j$

This means that $\sigma(\mathcal{C}) \subseteq \mathcal{B}_{\mathbb{R}}$.

2) $\mathcal{B}_{\mathbb{R}} \subseteq \sigma(\mathcal{C})$

Denote $\mathcal{C}' := \{(a,b) \mid a,b \in \mathbb{R}\}$

$$(a,b) = \left(\bigcup_{n=1}^{\infty} (-\infty, b-\tfrac{1}{n}]\right) \cap (-\infty, a]^c$$

$(-\infty, b-\frac{1}{n}] \in \sigma(\mathcal{C})$; $\bigcup_{n=1}^{\infty} (-\infty, b-\frac{1}{n}] \in \sigma(\mathcal{C})$; $(-\infty, a] \in \sigma(\mathcal{C})$; $(-\infty, a]^c \in \sigma(\mathcal{C})$; $(a,b) \in \sigma(\mathcal{C})$

$(a, \infty) = \bigcup_{n=1}^{\infty} (-\infty, a]^c \in \sigma(\mathcal{C})$

Thus $\mathcal{C}' \subseteq \sigma(\mathcal{C})$.

By the same logic as in 1), we have that $\mathcal{B}_{\mathbb{R}} \subseteq \sigma(\mathcal{C})$

Since $\sigma(\mathcal{C}) \subseteq \mathcal{B}_{\mathbb{R}}$ and $\mathcal{B}_{\mathbb{R}} \subseteq \sigma(\mathcal{C})$,

$$\sigma(\mathcal{C}) = \mathcal{B}_{\mathbb{R}}$$ ∎